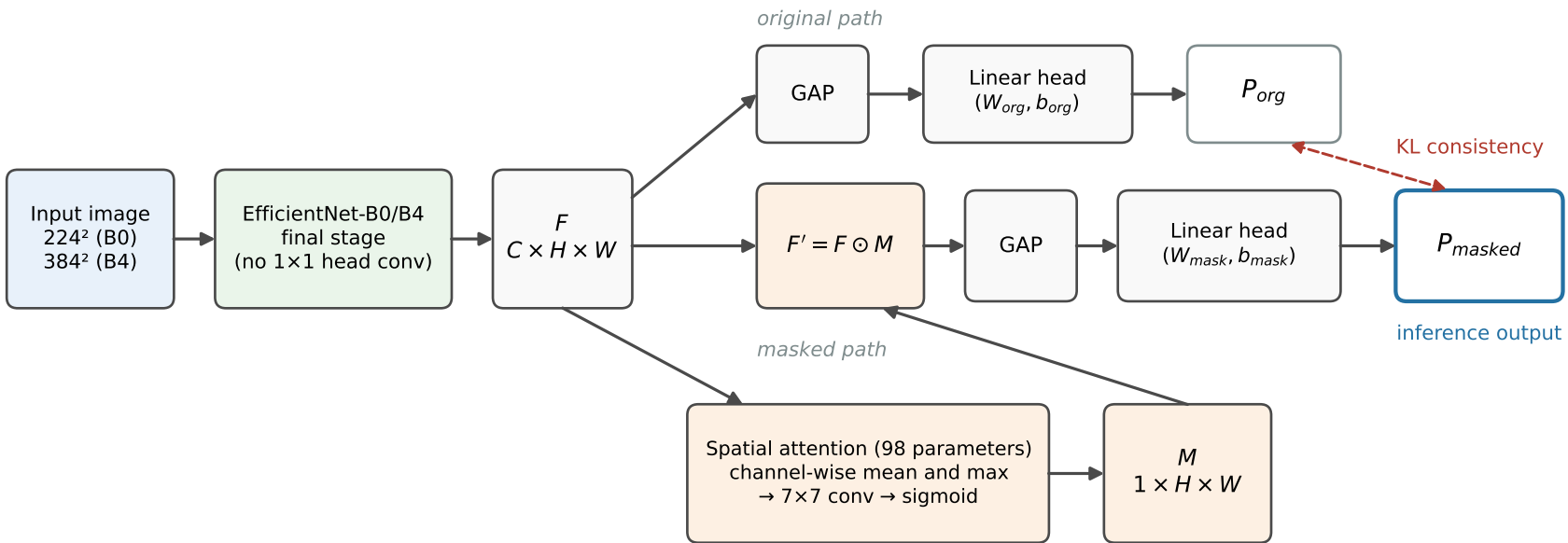


(a) SC-EfficientNet (Supervised-Consistency EfficientNet)



(b) Training objectives

KL-only ablation

$$\mathcal{L}_{KL-only} = \mathcal{L}_{cls}(P_{org}, y) + \lambda \text{KL}(\text{sg}(P_{org}) \parallel P_{masked})$$

M is trained only through the KL term and collapses to a nearly constant map (mean ≈ 0.98 – 0.99).

SC-EfficientNet (proposed)

$$\mathcal{L}_{SC} = \mathcal{L}_{cls}(P_{org}, y) + \mathcal{L}_{cls}(P_{masked}, y) + \lambda \text{KL}(\text{sg}(P_{org}) \parallel P_{masked})$$

$$+ \beta_{area} (\bar{m} - \rho)^2 + \beta_{bin} \text{mean}(M \odot (1 - M))$$

$\lambda = 0.10$, $\rho = 0.5$, $\beta_{area} = 1$, $\beta_{bin} = 0.1$; selective mask ($\approx 50\%$ of positions < 0.5)

Controls: standard baseline = full EfficientNet (with 1×1 head conv) → GAP → linear; headless = final-stage features → GAP → linear. sg = stop-gradient.